

2026 MEN310 Final Project

[Announcement] Final Project: AI-Driven Thermal Design via PINNs

Course Code: MEN310 Heat Transfer (Spring 2026)

Instructor: Prof. Jaeseon Lee

1. Project Overview & Objectives

The goal of this final project is to bridge the gap between classical heat transfer theory and modern AI-driven engineering. Students will select a physical heat transfer system and implement **Physics-Informed Neural Networks (PINNs)** to solve forward or inverse problems. Through this project, you will:

- Deepen your understanding of governing equations and boundary conditions.
- Gain hands-on experience in "Physics-ML" by embedding PDEs into neural networks.
- Develop critical engineering perspectives by verifying AI-generated outputs against physical intuition.

2. Technical Requirements & Methodology

- **Methodology:** All projects must be implemented using **PINNs (Physics-Informed Neural Networks)**. Standard data-only ML models or traditional FDM-only solvers will not be accepted.
- **Problem Complexity:** * To ensure educational value, **trivial problems** (e.g., simple 1D steady-state conduction without heat generation) are excluded.
 - Conversely, avoid overly complex 3D transient problems that require **high-end GPU resources** beyond standard desktop capabilities.
 - Your project must include at least one of the following: (1) Internal heat generation ($\dot{q}$), (2) Transient analysis, (3) Combined convection-conduction, or (4) Inverse parameter estimation (Inverse Problem).
- **PINN Libraries:** While basic PyTorch implementation is encouraged, using specialized PINN libraries such as **DeepXDE** will receive **extra credit (bonus points)** for demonstrating advanced tool utilization.

3. Project Phases & Deadlines

All submissions must be uploaded to the **Heat-PINN Space** platform by the specified deadlines.

Phase	Deliverable	Deadline	Content Requirements
Phase 1	Project Proposal	May 15, 18:00	Topic selection, mathematical modeling (PDE/BCs), and scaling strategy.
Phase 2	Midterm Report	May 27, 18:00	Source code, loss convergence plots, and an Ablation Study .
Phase 3	Final Report	June 12, 18:00	Final source code, visualization of results, and Physical Soundness analysis. 10 pages of your presentation slide

4. Submission Guidelines

- **Template:** All reports MUST follow the provided **official templates (.docx)** available on the blackboard.
- **Report Content (Phase 2 & 3):** You must include the full source code (or a link to a Colab/Jupyter Notebook), visual execution results, and your own critical analysis of the physics and AI performance. **Final submission includes your presentation slide.**
- **Language:** All project deliverables, including the source code comments and reports, must be written in **English**.

5. Evaluation Criteria

Your work will be evaluated based on:

1. **Physical Validity:** Does your model correctly reflect the heat transfer governing equations?
2. **Convergence Effort:** How did you address scaling, normalization, or gradient issues?
3. **Critical Thinking:** How did you verify the AI's results using your engineering knowledge? Did you identify and correct LLM "hallucinations"?
4. **Ablation Study:** Did you systematically analyze the impact of network depth, nodes, or activation functions?